08. reading and writing files

```
Data Sciences Institute
Python
```

Learning objectives

- Be able to open, read and write files
- Navigate folders and manipulate paths

Navigating folders

- Sometimes we need to work with many files at once, like reading all the files in a folder.
- Python has a built-in module called `os` that helps us look at and navigate the files and folders on our computer.
- With `os`, we can:
  - See which files and folders are in a folder
  - Check if a file or folder exists
  - Create new folders
  - Change the current working folder
- This lets us write programs that work with multiple files automatically, instead of opening them one by one.

Navigating folders

```
import os

# get path to the folder we're currently in
os.getcwd()

# see contents of the current folder
os.listdir()

# change to a new directory
os.chdir()
```

Navigating folders

```
# Path to the folder you want to look inside
folder_path = '../../05_src/data/slides_data'

# Get a list of all files and folders inside
items = os.listdir(folder_path)

# Print each item
for item in items:
    print(item)

delay_reasons.csv
all_delays.csv
bike_thefts_joined.csv
california_housing_test.csv
…
```

Opening, reading and writing files

- Python can read and write files using the built-in open() function.
- Syntax: `open(file_path, mode)`
  - file_path: string indicating the location of the file
  - mode: one-character string indicating how to open the file

Mode	Description
'r'	Read only. Produces an error if the file doesn't exist
'w'	Write. Creates a new file if it doesn't exist, or overwrites an existing file
'a'	Append. Adds to an existing file, or creates a new file if it doesn't already exist

Opening files

- To open a file, we use a with statement which follows the pattern
  - `with open(‘file path’) as variable_name`
- Then an indented block of code to process the file

```
# Example
with open('../../05_src/data/slides_data/
california_housing_test.csv', 'r') as file:
    print(file)

<_io.TextIOWrapper
name='../../05_src/data/slides_data/california_housi
ng_test.csv' mode='r' encoding='UTF-8'>
```

This tells you Python successfully opened the file.

Reading files

- As we saw, opening a file doesn't immediately get us the file's contents. To do that, we must use a read method

Method	Description
`read()`	returns the full file contents, which can be overwhelming for larger files
`readline()`	returns only the next line in the file. Python keeps track of where it is in the file.
`readlines()`	returns the full file as a list. Each item is one line in the file.

```
with open('../../05_src/data/slides_data/california_housing_test.csv', 'r')
as file:
    print(file.read())
```

This reads the entire file!

Reading files

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```
with open('../../05_src/data/slides_data/california_housing_test.csv', 'r')
as file:
      print(file.readline()) # First line
      print(file.readline()) # Second line
```

This reads the line one at a time

Reading files

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```
with open('../../05_src/data/slides_data/california_housing_test.csv','r')
as file:
      lines = file.readlines()
for line in lines:
      print(line)
```

This reads everything in the file as a list of lines!

Writing files

- There are also methods to write.

Method	Description
`write()`	Writes a single string to file
`writelines()`	Writes a list (or any iterable) of strings to a file. Need to separate new lines

```
airlines = [
"Code\tAirline\n",
"AAL\tAmerican Airlines\n",
"ABK\tAirbus Canada\n",
"ABP\tABS Jets\n",
"ACA\tAir Canada\n"
]

with open('airlines.txt', 'w') as file:
      file.writelines(airlines)
```

Recall the escape sequences!

Code	Airline
AAL	American Airlines
ABK	Airbus Canada
ABP	ABS Jets
ACA	Air Canada

Appending files

- There are also methods to add to an existing file (append).

```
with open('airlines.txt’, ‘a') as file:
      file.write(”ARQ\tAirico\n")
```

Code	Airline
AAL	American Airlines
ABK	Airbus Canada
ABP	ABS Jets
ACA	Air Canada
ARQ	**Airico**

We added a code and airline to our existing text file!

Quiz: Reading and writing files

1. **What does a file path refer to?**

A. The name of a variable
B. The location of a file on your computer
C. The contents of a file
D. It’s a python keyword

Quiz: Reading and writing files

2. Which of the following mode opens a file for reading only

A. ‘r’
B. ‘w’
C. ‘a’
D. None of these

Quiz: Reading and writing files

3. Which of the following mode adds text to an existing file without deleting its contents?

A. ‘r’
B. ‘w’
C. ‘a’
D. None of these

Quiz: Reading and writing files

4. What happens if you open a file in ‘w’ mode that already exists?

A. Python raises an error
B. The file is deleted
C. The file is overwritten
D. The file is opened in read-only mode

Quiz: Reading and writing files

5. What does this code do?

```
with open('file.txt’, ‘a') as file:
      file.write(”Hello! \n”)
```

A. Deletes the file

B. Overwrites the file

C. Adds Hello! to the end of the file

D. Reads the file

Quiz: Reading and writing files

6. **Complete the code so it writes “Hello, Python!” to a file called output.txt**

```
with open('output.txt', 'w') as file:
      file._____(”Hello, Python!”)
```

Exercise: Part 1

- Write a code that:
    1. Asks the user for their name
    2. Saves it to name.txt

Exercise: Part 2

- Write a code that:
    1. Asks the user for their age
    2. Append it to name.txt

Exercise: Part 3

- Write a code that:
    1. Open name.txt
    2. Read and print the contents